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Similarity Matching Methods for Images and Text

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I. Abstract

This study evaluates three methods for measuring similarity between images and text. Although Multimodal Artificial Intelligence has grown quickly, connecting visual and textual information remains a major challenge. We organize this project as a small research paper around the main question: "How much better are modern multimodal models (CLIP) than traditional feature engineering and linear projection for image-text similarity matching?"

We evaluate three methods on 1,000 image-caption pairs from the Flickr8k dataset: (1) a naive baseline that compares ResNet50 visual features with TF-IDF text features; (2) a learned linear projection that maps visual data into text space; and (3) Contrastive Language-Image Pretraining (CLIP), a modern model that puts both images and text in a shared space.

Our test results show that the naive method almost completely fails ($R@1 = 0.2\%$), while linear projection provides only a weak connection ($R@1 = 5.5\%$). In contrast, CLIP achieves high accuracy ($R@1 = 55.0\%$) and successfully handles both random and hard negative samples. Through visual search examples, heatmaps, and error analysis, we highlight the strengths of a shared embedding space while also exposing CLIP's remaining limitations, such as its focus on broad concepts rather than precise contextual details.

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II. Introduction

This research investigates the field of **Multimodal Artificial Intelligence**, with a focus on cross-modal retrieval and image-text matching. Over the past decade, unimodal AI models have achieved strong results. Computer vision models, such as ResNet [1], excel at extracting visual features, while Natural Language Processing (NLP) techniques, ranging from TF-IDF [2] to Transformer architectures [3], effectively capture text representations. However, the fundamental barrier in multimodal AI is the **Semantic Gap**: image vectors and text vectors naturally exist in different vector spaces. Directly calculating the similarity between them yields meaningless results.

To address this, we structure our project around a core research question:

How much better are modern multimodal embedding models (CLIP) than traditional feature engineering and linear projection for image-text similarity matching?

We evaluate three methods that show the development of AI: from handcrafted features, to basic learned mappings, and finally to deep multimodal learning. The main contributions of this paper are:

- Provide a clear baseline for comparing three generations of similarity matching techniques.
- Expand the evaluation metrics to include standard retrieval benchmarks such as Recall@1, Recall@5, and Mean Reciprocal Rank (MRR).
- Conduct detailed error analysis and targeted tests to show not only how well the models perform, but also *why* they succeed or fail.

III. Research Problem

The main research problem is **Image-Text Matching**: given an image and a caption, the system must estimate whether they describe the same content.

- **Input:** An image I and a candidate text caption T .
- **Output:** A similarity score $S \in [-1, 1]$ and a ranked list of captions for retrieval.
- **Research Objective:** To compare whether traditional feature engineering, shallow linear projection, or modern multimodal embeddings produce the best image-text alignment.
- **Evaluation Metrics:** cosine similarity, Recall@1, Recall@5, and Mean Reciprocal Rank (MRR).

Research Hypothesis:

We expect independent ResNet50 and TF-IDF features to fail because they belong to different vector spaces. We expect linear projection to improve retrieval slightly by learning statistical correlations, but still fail on hard negatives. Finally, we expect CLIP to perform best because it was trained directly to align images and text in a shared embedding space.

IV. Related Work

The challenge of aligning images and text has developed across three major phases:

- **Handcrafted Features:** Initially, researchers used traditional visual features (such as SIFT [4]) and text representations (such as Bag-of-Words and TF-IDF [2]). These separate representations were often aligned using Canonical Correlation Analysis (CCA) [5] to learn a shared projection space. Although these approaches worked for simple retrieval tasks, they struggled to capture high-level meaning and became weak when deeper understanding was required.
- **Early Deep Learning:** With the rise of deep neural networks, convolutional neural networks (CNNs) such as ResNet [1] and Transformer-based language models [3] allowed richer feature extraction. Researchers learned shallow mappings between independently trained visual and textual embeddings. This improved retrieval performance, but it often required task-specific training and did not generalize well. Although they were better than handcrafted approaches, these methods could be expensive to train from scratch and could overfit small datasets.
- **Foundation Models Era:** The field changed with OpenAI’s CLIP (Contrastive Language-Image Pretraining) [6]. By applying large-scale contrastive learning [7] to about 400 million internet image-text pairs, CLIP [6] learns to pull matching image-text pairs closer together while pushing non-matching pairs farther apart in a shared embedding space. This created a strong shared semantic space that became a standard for modern cross-modal tasks.

V. Methodology

We evaluate three approaches to image-text matching. For all methods, alignment is measured using cosine similarity, a standard similarity metric in information retrieval [8]:

$$\text{Cosine}(A, B) = \frac{A \cdot B}{\|A\| \times \|B\| + \epsilon}$$

Method 1: Naive Handcrafted Feature Alignment We independently extract visual features using a pre-trained ResNet50 model and text features using a TF-IDF vectorizer. We then compare them using cosine similarity. The main limitation is that the ResNet visual features and TF-IDF text vectors were created completely independently. Because they do not naturally exist in the same mathematical feature space, directly comparing them yields meaningless results.

Method 2: Learned Linear Projection To address the limitation of Method 1, we introduce a learned linear projection. Instead of comparing two unrelated feature spaces directly, we use a standard `LinearRegression` model to learn a mapping from the image features to the text features using paired training examples. Once the visual features are projected into the textual space, we can then apply cosine similarity to measure their alignment.

Method 3: Pretrained Multimodal Representation Learning (CLIP) Unlike the previous methods, CLIP was trained from the beginning to understand both images and text together. It is pretrained using large-scale contrastive learning [7] on hundreds of millions of image-text pairs [6]. During training, the model learns a shared embedding space where matching image-text pairs are pulled closer together while non-matching pairs are pushed farther apart. Because of this training goal, both image and text embeddings naturally exist in the same semantic space, so no additional mapping is required. We can directly compare them using cosine similarity for cross-modal retrieval.

VI. Experiments

VI.1. Dataset

This study uses the Flickr8k dataset [9], accessed through the Hugging Face `datasets` library via the `jxie/flickr8k` dataset repository [10]. It contains images of people and animals doing everyday activities, each paired with five descriptive captions. To keep the experiment efficient while ensuring reliable evaluation, we divide the dataset into two splits:

- **Training Set (6,000 images):** Used only to fit the TF-IDF vocabulary and train the Linear Regression mapping weights for Method 2.
- **Test Set (1,000 images):** Used to evaluate all three methods using different quantitative and qualitative metrics. We use the first valid caption for each image as the ground truth.

VI.2. Data Split

The experiment uses two dataset splits to separate model fitting from final evaluation.

- **Training Set:** 6,000 images are used to fit the TF-IDF vocabulary and train the Linear Regression mapping for Method 2.
- **Test Set:** 1,000 images are used to evaluate all three methods using similarity scores and retrieval metrics.

VI.3. Training Setup

Method 1 does not require training because it directly compares handcrafted feature vectors. Method 2 is trained with a Linear Regression model using ResNet50 image features as inputs and TF-IDF caption features as targets. Method 3 uses the pretrained `openai/clip-vit-base-patch32`

checkpoint, which has a visual encoder based on the Vision Transformer (ViT) architecture [11]. It does not need additional training.

Complexity Analysis Before analyzing performance, we consider the trade-off between model ability and computational cost for each method.

Method	Training Requirement	Inference Speed	Learnable Parameters
Naive Baseline	None	Very fast	0
Linear Projection	Seconds	Fast	≈ 1 million
CLIP	Fully pretrained	Medium	≈ 151 million

Table 1: Complexity and cost trade-offs for the three methods.

VI.4. Evaluation Setup

We conduct a quantitative evaluation by calculating standard retrieval metrics and similarity scores in three scenarios:

- **Positive Pair:** The cosine similarity between an image and its correct caption (testing matching quality).
- **Random Negative:** The cosine similarity between an image and a random caption (testing the ability to separate matches from non-matches).
- **Hard Negative:** A synthetic negative example where key subjects in the true caption are swapped (e.g., "dog" $\rightarrow$ "cat", "man" $\rightarrow$ "woman", "running" $\rightarrow$ "standing"). This scenario tests whether the model can handle captions that have surface-level word overlap.

We also evaluate global retrieval performance by ranking all 1,000 test captions against each query image. We report:

- **Recall@1 and Recall@5:** The percentage of cases where the true caption appears in the top 1 or top 5 ranked results.
- **Mean Reciprocal Rank (MRR):** The average of the reciprocal ranks of the correct caption, giving a broader measure of ranking quality.

VII. Results

Benchmark Table The quantitative results on the 1,000 test samples support our hypothesis that a shared multimodal space performs better.

Analysis

- **Method 1 (Naive Baseline):** This method performs no better than random guessing, with only 0.2% Recall@1. Its positive, random negative, and hard negative scores are all very close, showing that direct comparison between ResNet50 and TF-IDF features does not produce meaningful image-text alignment.

Method	Positive Pair ($\uparrow$)	Random Neg ($\downarrow$)	Hard Neg ($\downarrow$)	R@1	R@5	MRR
1 (Naive)	0.0340 ± 0.0434	0.0388 ± 0.0464	0.0329 ± 0.0435	0.002	0.005	0.0075
2 (Linear Proj.)	0.2016 ± 0.1307	0.0604 ± 0.0746	0.1635 ± 0.1145	0.055	0.170	0.1190
3 (CLIP)	0.3184 ± 0.0324	0.1704 ± 0.0452	0.2876 ± 0.0406	0.550	0.832	0.6736

Table 2: Comprehensive quantitative comparison on the Flickr8k test set.

- **Method 2 (Linear Projection):** This method improves the result to 5.5% Recall@1 and 17.0% Recall@5. It separates positive pairs from random negatives more clearly, but still struggles with hard negatives (0.1635). This suggests that the learned mapping captures shallow statistical overlap rather than deeper semantic meaning.
- **Method 3 (CLIP):** CLIP performs best, achieving 55.0% Recall@1 and 83.2% Recall@5. Its strong retrieval performance shows the advantage of contrastive pretraining and a shared multimodal embedding space.
- **Main Finding:** The results confirm that simply forcing independent image and text features together is not enough. Strong image-text matching requires representations that are trained jointly for cross-modal alignment.

VIII. Ablation Study

To check whether the linear projection method was limited by the TF-IDF vocabulary size, we conducted an ablation study with vocabulary sizes of 500, 1,000, and 2,000.

TF-IDF Vocab Size	Recall@1	Recall@5	MRR
500	0.055	0.170	0.1190
1000	0.056	0.193	0.1317
2000	0.055	0.198	0.1317

Table 3: Ablation study on Method 2 TF-IDF vocabulary size.

Increasing the vocabulary from 500 to 1,000 slightly improves Recall@5 and MRR, but expanding it further to 2,000 yields smaller improvements. This confirms that the main bottleneck for Method 2 is not vocabulary size, but the limited ability of a shallow linear layer to capture deep semantic mappings.

IX. Visualizations and Qualitative Retrieval

To make the model behavior easier to understand, we created several visual examples.

IX.1. Similarity Score Distribution

We plotted histograms showing the distribution of cosine similarity scores for positive pairs, random negatives, and hard negatives across the three methods. The distribution of similarity scores shows whether the model assigns higher scores to matching pairs and lower scores to non-matching pairs. While Method 1 and Method 2 struggle to maintain a clear boundary—especially when penalizing hard negatives—CLIP demonstrates a much clearer separation between positive pairs and negatives. This clear separation matches its better retrieval performance (Recall@K, MRR).

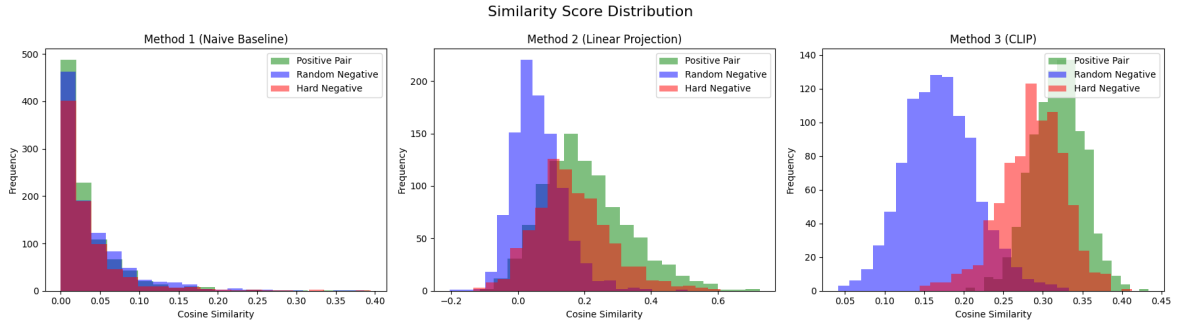


Figure 1: Similarity score distribution across positive pairs, random negatives, and hard negatives.

IX.2. CLIP Similarity Heatmap

We generated a cosine similarity matrix for 30 query images against 30 captions. A clear diagonal line appears with high similarity scores, showing that CLIP successfully pairs each corresponding i^{th} image with the i^{th} caption while giving lower scores to off-diagonal pairs.

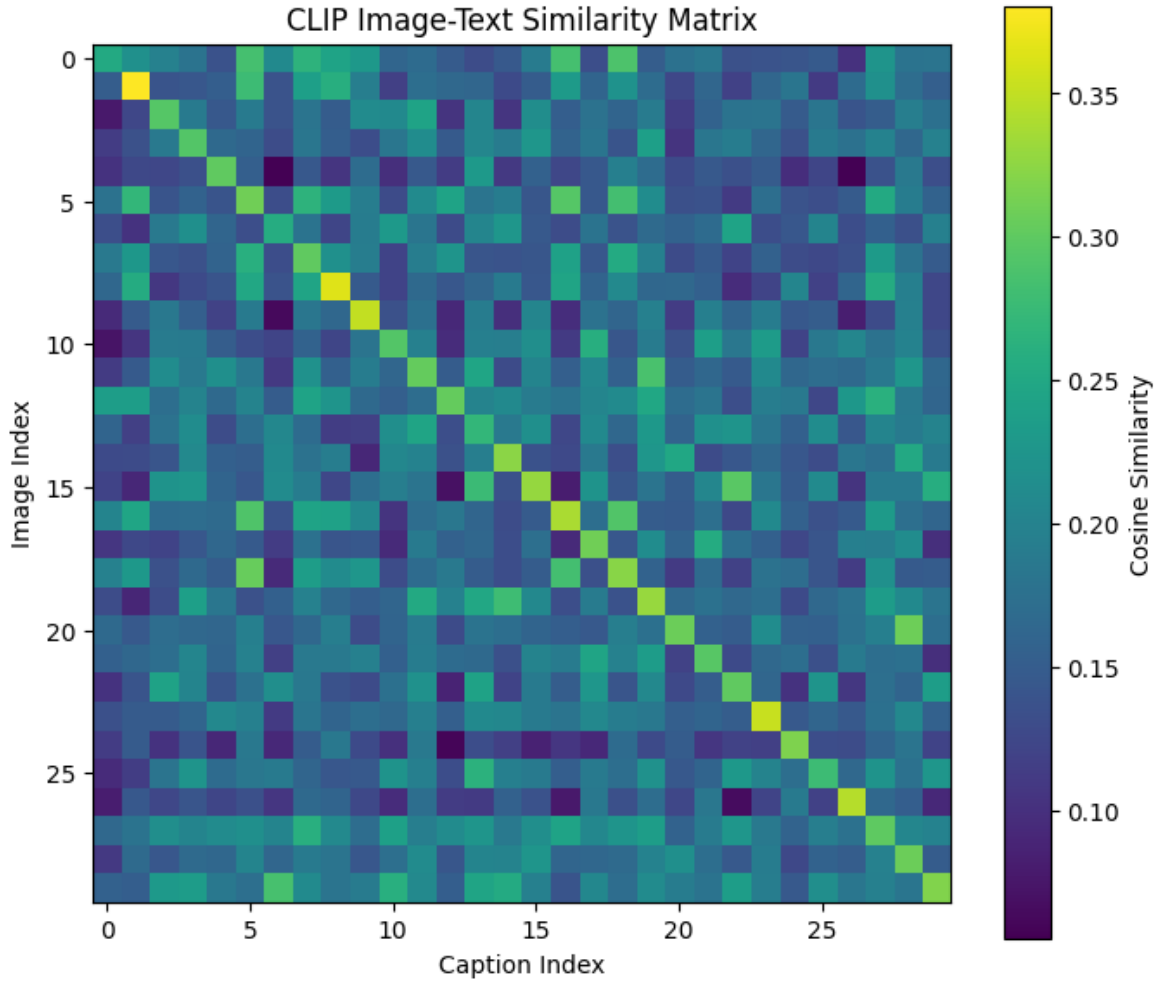


Figure 2: CLIP cosine similarity heatmap for query images and candidate captions.

IX.3. Qualitative Retrieval Demo

When CLIP searches the 1,000 test captions using a query image, the results are highly accurate. The following examples show the top retrieved captions for two query images from the test set.

Query Image Index 615 Ground Truth Caption: *"a group of woman in pink dresses walking across a crosswalk towards a cruissip ."*



Figure 3: Query image for qualitative retrieval demo, index 615.

Top 3 Retrieved Captions

1. Score=0.4334
a group of woman in pink dresses walking across a crosswalk towards a cruissip .
2. Score=0.3172
A group of women are performing onstage .
3. Score=0.3091
Girls dance in colorful outfits .

Query Image Index 779 Ground Truth Caption: *"A man does a trick on his skateboard in front of a crowd ."*



Figure 4: Query image for qualitative retrieval demo, index 779.

Top 3 Retrieved Captions

1. Score=0.3510
A man does a trick on his skateboard in front of a crowd .
2. Score=0.3347
A man is doing tricks on a bicycle on ramps in front of a crowd .
3. Score=0.3275
A man wearing a red helmet jumps up while riding a skateboard .

X. Discussion

The results clearly answer our research question: why does CLIP perform best?

- ResNet and TF-IDF fail because they are trained independently and produce representations in separate spaces.
- Linear projection fails because it maps inputs based on simple statistical correlations (co-occurrence of words) rather than real semantic alignment. This is clear in the hard negative test, where a synthetic noun change ("dog" to "cat") still gives a high similarity score because the surrounding context words remain the same.

- CLIP succeeds because its contrastive objective function explicitly forces the Vision and Text Transformers to learn a shared semantic representation space during training.

However, the hard negative scores for CLIP (0.2876) remain quite close to its positive pair scores (0.3184). Combined with our error analysis, this shows that CLIP is not immune to semantic shortcuts. It struggles with precise meaning combinations and can be distracted by dominant subjects while ignoring the surrounding context.

XI. Error Analysis and Limitations

Although CLIP performs very well, analyzing its failures reveals important details about its behavior.

Failure Case Analysis (Query Image 0):

- **Ground Truth (Rank 118):** *"The dogs are in the snow in front of a fence ."*
- **Rank 1 Retrieval (Score=0.3664):** *"Two Dobermans fighting or playing together on a lawn ."*
- **Rank 2 Retrieval (Score=0.3364):** *"A black and a blonde dog are either playing or fighting with each other ."*
- **Rank 3 Retrieval (Score=0.3292):** *"A black dog and a tan dog fighting ."*



Figure 5: Failure case for CLIP retrieval on query image 0.

Why did this fail? Although the true image shows dogs playing in the snow, CLIP retrieved captions about dogs playing on a lawn or fighting. This suggests that CLIP focuses strongly on the main visual subjects (the dogs and their active poses) but sometimes ignores important background details such as "snow" or "fence." This behavior is similar to a more advanced "bag-of-words" pattern: CLIP detects the strong presence of "dogs" and "playing", which matters more than the missing environmental details.

XII. Reproducibility

The project notebook provides the full experimental pipeline, including dataset loading, feature extraction, model training, retrieval evaluation, and visualization generation.

Execution Commands:

```
# Run the notebook or convert it to a Python script
jupyter notebook AIO2026_CONQUER_MODULE_1.ipynb
```

The notebook reports Recall@K metrics, Mean Reciprocal Rank, similarity tables, heatmaps, and qualitative retrieval examples needed to reproduce the analysis.

XIII. Conclusion and Future Work

By treating this project as an experiment-based study, we built a clear testing framework for evaluating three generations of image-text matching techniques. We showed that forcing two independent data spaces together with a shallow linear mapping creates only weak alignment. In contrast, modern foundation models such as CLIP rely on joint contrastive pretraining to reduce the Semantic Gap, achieving 55% Recall@1 on a 1,000-sample search space.

For future work, this method can be extended by testing newer multimodal foundation models such as SigLIP [12], adding more challenging spatial hard negatives, and using embeddings in vector databases such as Faiss to build scalable real-time multimodal search systems.

XIV. References

- [1] K. He et al., “Deep residual learning for image recognition”, in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 770–778. DOI: 10.1109/CVPR.2016.90
- [2] G. Salton et al., “Term-weighting approaches in automatic text retrieval”, *Information Processing & Management*, vol. 24, no. 5, pp. 513–523, 1988. DOI: 10.1016/0306-4573(88)90021-0
- [3] A. Vaswani et al., “Attention is all you need”, in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017.
- [4] D. G. Lowe, “Distinctive image features from scale-invariant keypoints”, *International Journal of Computer Vision*, vol. 60, no. 2, pp. 91–110, 2004. DOI: 10.1023/B:VISI.0000029664.99615.94
- [5] H. Hotelling, “Relations between two sets of variates”, *Biometrika*, vol. 28, no. 3/4, pp. 321–377, 1936. DOI: 10.2307/2333955
- [6] A. Radford et al., “Learning transferable visual models from natural language supervision”, in *Proceedings of the 38th International Conference on Machine Learning (ICML)*, vol. 139, 2021, pp. 8748–8763.
- [7] T. Chen et al., “A simple framework for contrastive learning of visual representations”, in *Proceedings of the 37th International Conference on Machine Learning (ICML)*, vol. 119, 2020, pp. 1597–1607.
- [8] C. D. Manning et al., *Introduction to Information Retrieval*. Cambridge University Press, 2008, ISBN: 9780521865715.
- [9] M. Hodosh et al., “Framing image description as a ranking task: Data, models and evaluation metrics”, *Journal of Artificial Intelligence Research*, vol. 47, pp. 853–899, 2013. DOI: 10.1613/jair.3994
- [10] Jxie, *Flickr8k dataset*, <https://huggingface.co/datasets/jxie/flickr8k>, Accessed: 2026-07-03, 2023.
- [11] A. Dosovitskiy et al., “An image is worth 16x16 words: Transformers for image recognition at scale”, in *International Conference on Learning Representations (ICLR)*, 2021.
- [12] X. Zhai et al., “Sigmoid loss for language image pre-training”, in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023, pp. 11 975–11 986.

Appendices

1. **GitHub Repository:** The full project code is available at [AIO2026 Conquer Module 01 GitHub repository](#).
2. **Google Colab Notebook:** The executable notebook is available at [AIO2026 Conquer Module 01 Colab notebook](#).